System Modeling and Prototyping

IE2044

Final Assignment

The Project Brief

Design and build a 2-digit digital thermometer that displays temperatures from 0°C to 99°C. The system must use an **LM35** precision temperature sensor and an **ICL7107** (A/D Converter and Display Driver). The output must be shown on two 7-segment displays. The final product should be PCB-mounted finished working circuit.

Phase I: Proteus Simulation (Voltage Scaling)

- **Sensor Logic:** The LM35 outputs 10mV per degree Celsius. At 25°C, it outputs 250mV.
- **The Converter:** Use the **ICL7107** in Proteus. It is a classic "Voltmeter" chip that converts an analog voltage into a 7-segment drive signal.
- **Reference Voltage:** Must use a Potentiometer to set the Reference Voltage (Vref) of the IC to exactly 100mV so that 10mV from the sensor equals "1" on the display.
- **Verification:** Adjust the LM35 temperature in Proteus and ensure the two 7-segment displays match the temperature value.

Phase II: EasyEDA Pro PCB Requirements

- **Component Selection: LM35 sensor:** Through-hole (so it can stick out of the case).
    - **ICL7107:** DIP-40 package (large but easy to solder).
    - **Resistors/Caps:** SMD 0805.

      (Feel free to use another package, if it is not available)

- **Layout Constraints:**
    - **Analog Grounding:** Keep the LM35 ground and the ICL7107 analog ground pins connected at a single "Star Point" to prevent display noise from jittering the temperature reading.
    - **Display Alignment:** The two 7-segment displays must be perfectly aligned horizontally.

- **Power:** The ICL7107 requires a dual power supply (+5V and -5V).

Assignment Guidelines

1. Group Composition: 2 students per group and Equal contributions in simulation, and report writing.
2. Deadline for the report: 27th April 2026
3. File Naming Convention: Final report must follow: IE2044_Assignment_Group_X.pdf

Report Format

1. Cover Page

    Course / Module name

    Lab Assignment Title:

    Group Number:

    Student IDs and Names:

2. Introduction
3. Design Methodology
4. Component Selection
5. Results & Testing
6. Discussion
7. Conclusion
8. References